

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Mon, Jul 13 · 6:00 AM – 10:00 AM** | PDOP **0.99 · 34 sv** | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

23.9 TECU

+0.3 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork



GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 92°F | Low 60°F | Wind max 12 mph | Gusts 14 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO — GOOD CONDITIONS

Partly cloudy | Wind: 11.6 mph (gusts 14.1 mph) | Precip: 0" | Temp: 59.6°–92°F



DRONE WIND ANALYSIS



GO Peak: 11.6/14.1 mph S

* Sunrise: 5:56 AM | Sunset: 8:33 PM | fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	1/3 mph		ESE
6 AM	1/2 mph		NW
7 AM	1/1 mph		WNW
8 AM	2/3 mph		SSE
9 AM	3/4 mph		S
10 AM	3/4 mph		SSE
11 AM	6/7 mph		S
12 PM	8/10 mph		S
1 PM	8/9 mph		S
2 PM	9/11 mph		S
3 PM	9/11 mph		S
4 PM	12/14 mph		S
5 PM	8/13 mph		SSW
6 PM	10/12 mph		S
7 PM	9/11 mph		SSE
8 PM	5/8 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: N/A

SITE: Napa County, CA (38.3459, -122.3047)

DATE: Monday, July 13, 2026

REPORT GENERATED: Monday, July 13, 2026

SURVEY VERDICT: GO

Clear geometry (PDOP 0.94–0.99), manageable ionospheric activity (Kp 4.7, TEC +0.3 TECU nominal), and favorable weather support field operations. Proceed as planned.

BEST SURVEY WINDOW

Monday, July 13 — 06:00 AM to 10:00 AM local time (PST)

PDOP 0.94 at peak; 34-satellite constellation available (multi-constellation); low ionospheric delay risk. Overcast conditions do not degrade GNSS performance. Post-10:00 AM PDOP degradation makes early-morning window optimal.

BEST 4-HOUR STATIC GNSS WINDOW

06:00 AM – 10:00 AM (Monday, July 13)

Sustained PDOP ≤ 0.99 across full window. Moderate geomagnetic activity (Kp 4.7 avg) warrants standard static occupation duration (≥ 45 min per setup); ionospheric delay is acceptable and will not require extended sessions if multipath mitigation is applied. Elevation mask 13° recommended based on terrain.

DRONE OPERATIONS ASSESSMENT

Favorable. Wind max 12 mph is well within safe operating limits. Overcast skies acceptable. No thermal turbulence expected. Visual line-of-sight (VLOS) operations unrestricted. Ground-based RTK or PPK feasible using multi-constellation corrections during the 06:00–10:00 AM window (PDOP optimal).

FIELD SAFETY WARNINGS

No wildfire activity detected in proximity. AQI 45 (Good) — no respiratory or visibility concerns. Standard field precautions apply. GPS PRN 11 is currently unhealthy (UNUSUFN); receivers will auto-exclude. This is already reflected in constellation count (34 satellites). No action required.

SUMMARY

Field operations are approved for Monday, July 13. Optimal survey window is 06:00–10:00 AM when PDOP = 0.94–0.99 with 34 satellites across GPS/GLONASS/Galileo/BeiDou. Ionospheric activity is moderate (Kp 4.7, TEC +0.3 TECU); static sessions should use standard 45-minute occupations with emphasis on multipath mitigation at the site (dwarf scrub, low sky obstruction). Weather is overcast but dry; wind 12 mph maximum. No fire or air quality hazards. Proceed with planned schedule.

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

Download .txt File

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.22 ft (-18 ft)	466 ft	788K af	81%
Lake Oroville	93.4 mi	861.2 ft (-39 ft)	900 ft	2,859K af	81%
New Melones Lake	100.1 mi	1,020.22 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Coast Range Alluvium and Sandstone (N100CACSTL) is a Quaternary and Tertiary mix of alluvial gravels, sands, and silts overlying older sandstone and shale bedrock, deposited by ancient stream systems and marine processes in the Napa Valley. Water availability at this site is moderate to good, with water table depths ranging from roughly 10 to 88 feet below ground surface, indicating the aquifer receives recharge from local precipitation and upland flow. The shallow portions (under 100 ft) are typically productive for domestic and small agricultural supply.

FORMATION

Name	Coast Range Alluvium and Sandstone
Age / Rock	Quaternary to Tertiary age; alluvial sand, gravel, silt, and fine sandstone.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	60–120 ft (6–12 floors)
Recharge Source	Precipitation on valley floor and upland outcrops; lateral flow from adjacent bedrock highlands
Transit Time	15–40 years to 100 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.8–7.5 (est. USGS/DWR)
Hardness	est. 200–400 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 300–600 mg/L (est. USGS/DWR)

Notes	No sample data — typical for formation; modest hardness and TDS expected for alluvial aquifer in Napa Valley.
TREND	
Trend	Declining long-term; shallowest readings 1977 (9.90–54.10 ft), deepest 2014–2022 (21.86–87.71 ft), indicating multi-decadal drawdown consistent with regional groundwater extraction.
Source	USGS NWIS historical readings at site (1977, 2014, 2022)
Seasonal Swing	est. 10–25 ft seasonal swing (est. USGS/DWR)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The North Coast Ranges near 38.3459, -122.3047 are underlain by the N100CACSTL aquifer system, characterized by well-drained soils and consolidated bedrock with limited clay interbeds. This area lies outside any critically overdrafted groundwater basin, indicating historical groundwater extraction has remained within sustainable limits and has not induced significant aquifer-system compaction. For surveyors operating in this region, published survey benchmarks and monumentation are expected to remain stable, and subsidence is not a primary concern affecting vertical control.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<0.5 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This marine climate has mild winters and cool summers with persistent coastal winds, so your envelope needs robust air sealing and wind-resistant details rather than heavy insulation—focus on airtightness and protecting against the 92 mph wind loads with properly braced walls and secured roofing. Maximize south and west-facing glazing for daylighting and occasional winter warmth, but use deep overhangs or exterior shading on west-facing glass to block summer sun and reduce cooling loads, while minimizing north-facing glass to avoid heat loss on the cool side of the building. The key passive move here is natural ventilation through operable windows—design your plan so prevailing coastal breezes can cross-ventilate living spaces during the mild shoulder seasons, dramatically reducing mechanical cooling needs and taking advantage of this climate's greatest asset.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal marine site presents well-drained gravelly loam soils suitable for conventional spread footing design. Seismic parameters are currently unavailable and may warrant geotechnical investigation to confirm Site Class D and establish design accelerations. Wind exposure and moderate snow loads suggest standard structural detailing, though marine coastal conditions may influence material durability considerations.	
FOUNDATION	
Type	Conventional spread footing anticipated
Basis	Well-drained Coombs gravelly loam with 2-5% slopes suggests adequate bearing capacity for typical structures. Consult a licensed engineer to confirm bearing capacity and settlement estimates based on site-specific investigation.
STRUCTURAL IMPLICATIONS	
Seismic	Seismic design parameters (Ss, S1, SDS, SD1) are currently null and may warrant confirmation. Site Class D assumed for planning purposes. Consult a structural engineer to establish appropriate seismic design category and detailing requirements once parameters are available.

Wind Exposure	Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II, 92 mph 3-second gust wind speed. Verify final exposure category and terrain characteristics with site assessment.
Wind	Standard wind load design anticipated for Risk Category II. No hurricane or extreme wind zone exposure suggested for this inland coastal location; consult engineer to confirm final design wind pressures.
Snow	Moderate snow load of 5.8 psf ground snow anticipated with W2=0.35 winter wind factor. Standard roof design likely adequate; consult engineer for roof configuration and snow drift evaluation if applicable.
INVESTIGATIONS & INSPECTIONS	
Geotech Required	Yes - geotechnical investigation recommended to confirm Site Class D, establish seismic parameters, verify bearing capacity, and assess settlement potential for the Coombs gravelly loam soils.
Liquefaction Screen	Liquefaction screening recommended once seismic parameters are confirmed and shallow groundwater conditions are assessed. Well-drained gravelly soils suggest lower liquefaction potential, but site-specific evaluation warranted.
Special Inspections	Special inspections anticipated once seismic design category is established. Consult engineer and building official for final requirements based on confirmed SDC.
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes. Geotechnical investigation strongly recommended to confirm soil profile, confirm Site Class designation per ASCE 7-22 Section 20.3, and obtain seismic design parameters for this location.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial deposits on gently rolling foothills and valley margins of the California Coast Ranges. The soil exhibits a gravelly surface layer over a clay loam argillic horizon, typical of Mediterranean climates with winter moisture and summer drought. Site visitors should expect firm, stable ground with moderate gravel content and good load-bearing capacity, though subsurface clay requires attention during grading and utility installation.

Alluvial and colluvial parent material on low-angle slopes; mixed oak-grass and chaparral vegetation typical of well-drained upland benches.

DRAINAGE	
Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None
ENGINEERING	
Bearing Capacity	3,500–4,500 psf (shallow foundations on gravelly A horizon)
Rating	Good
Shrink-Swell	Moderate — clay loam argillic horizon may exhibit seasonal movement
Frost Action	Low
Cut Suitability	Good (gravelly upper layers provide stable excavation material)
Fill Suitability	Fair (clay substratum requires moisture control and proper compaction)
SEPTIC	
Suitability	Conditional
Notes	Percolation test required — clay loam B horizon (15–25 inches) may restrict infiltration; depth to bedrock and groundwater should be verified
HAZARDS	
Excavation	Moderate — 20–35% gravel content may increase excavation costs and equipment wear
Expansive Clay	Low — clay fraction typically non-expansive, though monitoring recommended
Erosion	Low on slopes under 5%; moderate risk in disturbed areas and cut banks; stabilize during construction
Other	None anticipated

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdnldataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 13, 2026, 1:14 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil well located 7.2 miles away, which is well beyond any actionable distance threshold for foundation review, surface easements, or environmental assessment. The remaining seven wells in the broader 10-mile radius pose no site planning constraints.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Site Planning Assessment The project site has three active mineral producers within 5 miles that warrant attention: Basalt Rock Co (2 miles, clay operations), Mt George Pumice Deposit (3.1 miles, pumice producer), and Silva Pumice Quarry (5 miles, pumice producer). Expect heavy truck traffic on access roads during operating hours from these facilities, with potential dust generation from aggregate and clay operations and noise from crushing or processing activities at the pumice sites. No inactive deposits are located within 1 mile to trigger Phase 1 Environmental Site Assessment consideration.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Planning Assessment Summary **Project Location:** 38.3459, -122.3047 (Solano County region) Federal permitting or funding will trigger ESA Section 7 consultation given the 12 species with critical habitat within 25 miles. The most planning-relevant species for terrestrial development are **California tiger salamander** and **Contra Costa goldfields** (habitat-specific constraints), while **California red-legged frog** and **vernal pool fairy shrimp** represent wetland/seasonal pool dependencies that commonly drive site layout modifications. The recommended next step is a Phase I biological assessment to map vernal pool and grassland communities on-site, establish species presence/absence, and quantify habitat loss; this will determine whether formal Section 7 consultation and incidental take authorization are required before project approval. Early coordination with USFWS will identify whether habitat mitigation, avoidance measures, or conservation easements can facilitate project viability. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment Summary This site in Sonoma County supports a **mixed oak woodland and chaparral community**, dominated by blue oak and California black oak with scattered California bay and buckeye, underlain by herbaceous and scrub understory species typical of Interior Coast Ranges foothills. The ecology reflects warm, dry summers and cool, wet winters characteristic of Mediterranean-influenced terrain, with soils that are likely well-drained but seasonally moist in lower areas—critical considerations for grading since compaction and drainage alterations could severely impact both the oak recruitment and spring-ephemeral wildflower populations evident in the observations. The presence of moisture-loving species like Braun's Giant Horsetail and California Wild Rose alongside drought-tolerant oaks and mariposa lilies indicates variable soil moisture zones across the site, suggesting careful drainage design will be needed to preserve seasonal water availability without creating erosion or saturation problems. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
<i>Crotalus oreganus oreganus</i>	635 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
<i>Latrodectus hesperus</i>	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
<i>Dasymutilla aureola</i>	91 obs
<i>Dasymutilla californica</i>	91 obs
<i>Dasymutilla sackenii</i>	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs

SPECIES	OBSERVATIONS
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU
 Field Safety Briefing: Napa County Area (38.3459, -122.3047) **Highest-Priority Hazards:** Northern Pacific rattlesnakes are your dominant concern with 635 documented observations—stay alert in rocky outcrops, grassland, and brush where they shelter. Poison oak is ubiquitous (1439 observations across multiple toxic plant species) and causes severe allergic reactions; it thrives year-round in this region. Western black widow spiders and various wasp/velvet ant species present secondary but manageable risks. **Practical Avoidance:** Wear closed-toe boots and long pants; stomp or tap the ground ahead when moving through brush to alert snakes. Learn poison oak by sight (three leaflets, variable coloring) and avoid touching all suspicious vegetation. Tuck pants into socks if moving through dense growth. Carry a first-aid kit rated for snake bite and have emergency contacts ready. **Seasonal Note:** Rattlesnakes are most active April–October when temperatures warm; exercise heightened vigilance during peak survey season. Poison oak remains a hazard year-round but is easier to identify when bare in winter months. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8

Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200

Total Observations

90

Total Species

65

Birds

5

Mammals

14

Reptiles

6

Amphibians

BIRDS (65 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3
Canada Goose	<i>Branta canadensis</i>	3

+ 50 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2

COMMON NAME	Scientific Name	OBS
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Western Toad	<i>Anaxyrus boreas</i>	2
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a diverse wildlife community characteristic of oak woodland and grassland habitats with adjacent riparian or wetland corridors, as evidenced by the presence of both upland species (California Quail, Western Bluebird, Mule Deer) and obligate aquatic/semi-aquatic fauna (Snowy and Great Egrets, Northwestern Pond Turtle, Pacific Chorus Frogs). Several species of regulatory concern are present, including the California-listed Foothill Yellow-legged Frog and Western Toad, which indicate accessible seasonal water features and relatively intact breeding habitat; the presence of these sensitive amphibians suggests areas of cool, clean water with minimal chemical disturbance. The diverse reptile assemblage, particularly the abundance of native snake species, indicates moderate to good habitat quality with adequate structural complexity, though the presence of non-native species (Eastern Fox Squirrel, American Bullfrog, Pond Slider) and relatively high bird diversity suggests the site experiences some human modification and may serve as wildlife corridor within a partially developed landscape. Development activities should avoid impacts to seasonal wetlands, riparian zones, and upland refugia that support this sensitive amphibian and reptile community. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

* PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↪ excellent
7/12	30.3°	98.8%	↪ excellent
8/12	33.7°	99.3%	↪ excellent
9/12	36.9°	99.8%	↪ annual optimal
10/12	39.8°	99.8%	↪ excellent
12/12	45°	99%	↪ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔗 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Story You're in a sweet spot for solar—with about 6.6 peak sun hours per day on average, your panels will generate solid power year-round, though expect nearly twice as much production in summer (9 hours) as in winter (3.9 hours), which actually matches when most people need cooling help. Your roof's south-facing slope of about 38 degrees is nearly perfect for capturing that sun, but here's the thing most people miss: if your roof is dark, it's absorbing three times more heat than a light-colored roof would—that's like choosing between black and white car interiors on a hot day, and it forces your air conditioning to work much harder. The real problem child is your west-facing windows, especially in summer afternoons, when they're pouring in twice the heat of your south windows; even with good blinds, that afternoon sun can spike your cooling costs noticeably. Plant a leafy deciduous tree on your south side if you can—it'll block summer sun when you need it most (4 hours of afternoon shade), then lose its leaves in winter to let warmth in when the days are short; it's like free seasonal insulation. Thermal mass (think of your walls and floor acting like a battery that slowly releases heat) matters more with a super-insulated home like ICF construction, which can hold warmth for 7 hours, whereas a standard frame home only holds it for about an hour. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing: 38.34586°N, 122.30466°W, Napa County

Structural Design Drivers

Seismic design controls the structural envelope for this site. Wind and snow load data are not available; however, the site location in Napa County places it within a known seismic region—the data explicitly confirms no Quaternary faults within the search radius, indicating faults have been evaluated and ruled out as proximate hazards, but seismic loading remains a code-driven requirement. IECC Climate Zone 3C (Marine) with accumulation factor index of 0°F-days eliminates snow and freeze-thaw cycling as structural considerations. Without wind speed data, assume current IBC or ASCE 7 seismic design criteria govern lateral load path, damping strategy, and foundation anchorage. Confirm current seismic design category (SDC) through Napa County building department and USGS seismic hazard maps before finalizing structural system.

Geotechnical Foundation Design

The site is underlain by Coombs gravelly loam, 2 to 5 percent slopes, classified as Hydro Group C with well-drained characteristics. Hydro Group C soils are moderately permeable; well-drained gravelly loam supports conventional shallow foundations without dewatering complications. No frost depth data is provided—this is critical for footing design and must be confirmed through local geotechnical authority or the Natural Resources Conservation Service for Napa County. The absence of subsidence basin data and Quaternary fault activity indicates no special mitigation for ground settlement or liquefaction is evident from regional databases; however, site-specific geotechnical investigation is mandatory to confirm bearing capacity, moisture content, and settlement potential. Texture classification is incomplete in the dataset—obtain full soil profile and engineering properties (CBR, Atterberg limits, angle of internal friction) before foundation depth and allowable bearing pressure are finalized.

Hazard Assessment

No Quaternary faults are mapped within the search radius, and no subsidence basin data exists for this location. These negative findings reduce (but do not eliminate) ground failure risk. Wildfire data are not provided in the briefing package; however, Napa County has documented fire activity—consult CAL FIRE Fire and Resource Assessment Program (FRAP) data and Napa County emergency management for defensible space and structure hardening requirements specific to this parcel. Seismic risk remains the primary geologic hazard and must be addressed through standard code-compliant lateral load design.

Site Context: Mining and Climate

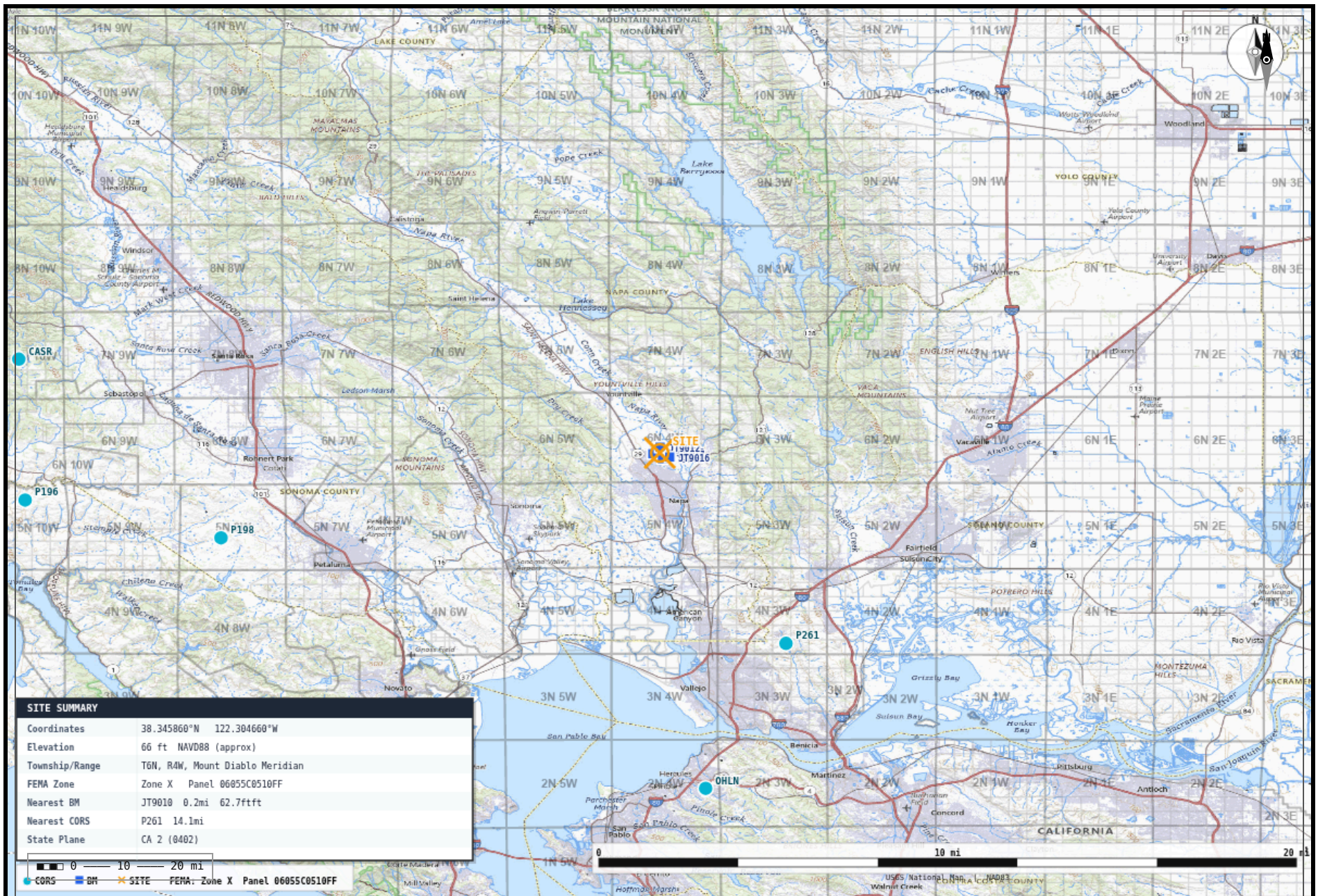
Three active mineral extraction operations are located within 5 miles: Basalt Rock Co (2 miles, clay); Mt George Pumice Deposit (3.1 miles, pumice); and Silva Pumice Quarry (5 miles, pumice). These operations may generate ground vibration, dust, and traffic. Request blast logs and operational schedules from county planning or mineral resource management; confirm whether vibration limits apply to structural design. No active oil or gas wells are producing within 10 miles (nearest Mobil well is 7.2 miles, plugged and abandoned). The IECC Zone 3C (Marine) classification with 0°F-days accumulation confirms minimal thermal mass, moisture control, and wind-driven rain are the envelope design drivers for any enclosure system.

Next Steps

Retain a licensed California geotechnical engineer to perform Phase 1 site assessment, confirm frost depth, complete soil profile characterization, and quantify foundation bearing capacity before structural system selection proceeds.

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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

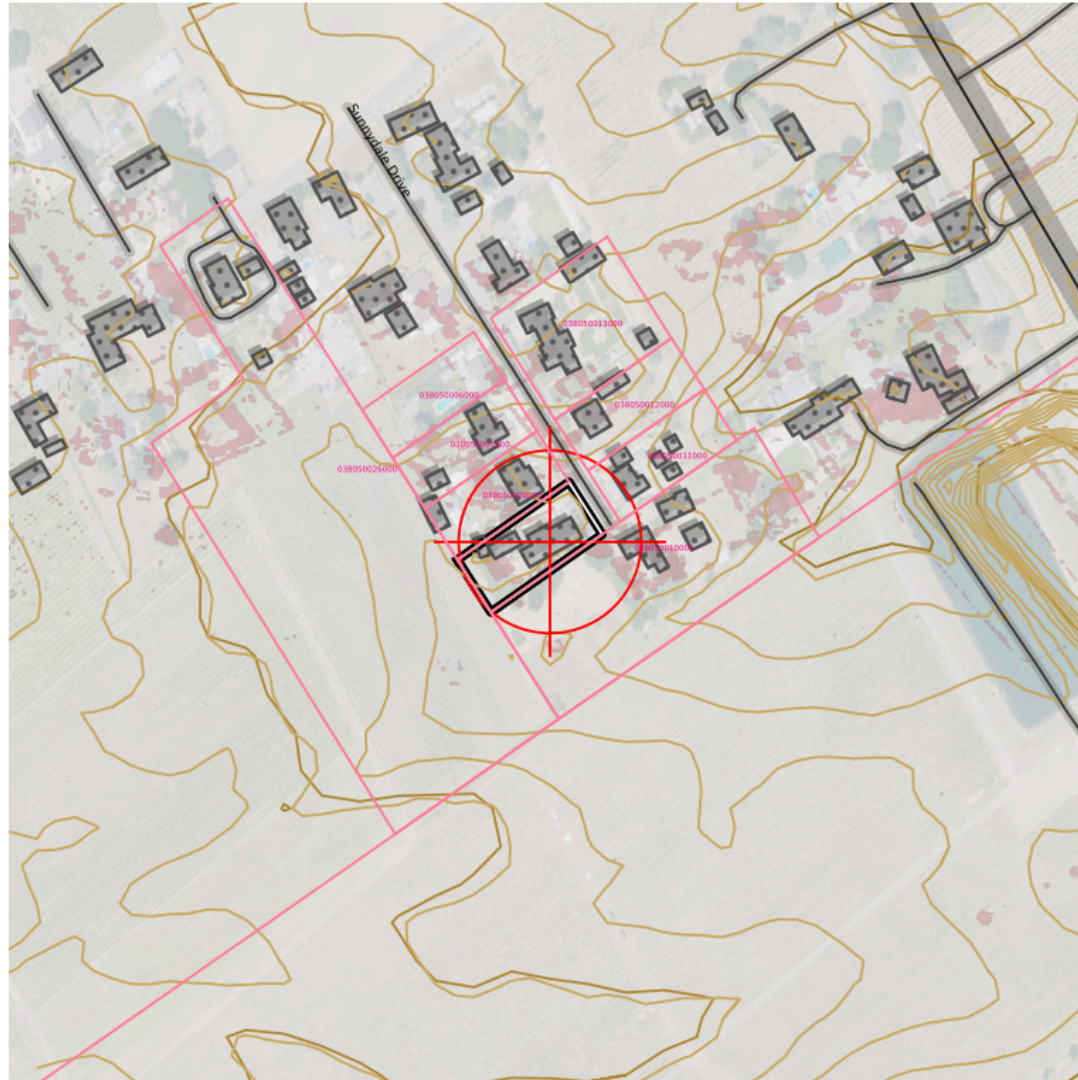


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.34586°N 122.30466°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

